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Data Article

ChatgaiyyaAlap: A dataset for conversion from Chittagonian dialect to standard Bangla

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A B S T R A C T

Bangla is one of the most used languages around the world with 240 million speakers. The standard Bangla language is the official language of people from Bangladesh and a few other parts outside Bangladesh, like West Bengal and Tripura. Although, people from different areas of Bangladesh do not use standard Bangla on a day-to-day basis. Instead, dialects of the Bangla language are used. The dialects of the Bangla language are quite diverse which includes the Rajshahi dialect, Sylheti dialect, Old Dhaka dialect, Chittagonian dialect, and many more. Nearly every division of Bangladesh has its unique dialect which adds linguistic diversity to the language. The main difference between the standard Bangla language and Chittagonian dialect is that it does not have any written form and the words vary from the standard Bangla. We built a dataset containing standard Bangla and one of its most used dialects: the Chittagonian dialect. Our dataset, "ChatgaiyyaAlap," was created by combining 4012 unique sentences in standard Bangla with their Chittagonian translations, which were gathered from dramas, comments, and videos on YouTube and Facebook. In our dataset, we cleaned the data by removing emojis, punctuations, and unessential

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spaces. The dissimilarity in spelling as well as in sentence structure between standard Bangla and Chittagonian dialects, particularly in negative sentences can be clearly visible through this dataset. Additionally, to maintain the data accuracy, we developed a dictionary containing 1,500 standard Bangla words and their Chittagonian form. The dictionary showcases significant variation in the vocabulary of both languages and resolves ambiguities along with potential biases. To perform a word-to-word translation this dictionary would be useful, while for studying comparative linguistics and developing intelligent systems the ChatgaiyyaAlap dataset might assist the researchers. Moreover, this dataset can be utilized to train language models for linguistic translation.

Specifications Table

Subject	Computer Science Applications, Artificial Intelligence, Linguistics
Specific subject area	A Comprehensive Dataset on the Bangla Language and Chittagonian Dialect, facilitating the exploration of linguistic variations.
Type of data	Comma Separated Value (.CSV)
Data collection	Data was collected from various social media platforms. A large number of data samples were taken from videos, posts, and comments available on different platforms. The files have been transformed into the .CSV format after removing punctuation, emojis, and spelling mistakes. Besides, the dictionary was helpful to avoid data inaccuracy.
Data source location	The data was gathered from social media, a mix of posts, comments, and videos.
Data accessibility	Repository name: Mendeley Data Data identification number: 10.17632/wtms9xbkkw.1 Direct URL to data: https://data.mendeley.com/datasets/wtms9xbkkw/1 Instructions for accessing these data:
Related research article	N/A

1. Value of the Data

- Developing a clean and distinct dialect-based dataset for low-resource language using various resources is challenging work to be done. Utilizing this affluent dataset researchers can convert Chittagonian dialect into standard Bangla using different machine learning and deep learning models.
- Every language has its discrete sentence pattern, and this dataset can assist researchers in understanding the complex forms of the Chittagonian dialect.
- A well-structured dataset was formed as a result of the large number of data-collecting sources used, including plays, films, and comments in the Chittagonian dialect that were posted on Facebook and YouTube. Therefore, researchers can use the data to convert the Chittagonian dialect or Bangla language into any other language.
- Researchers may utilize this dataset to translate the Chittagonian dialect into any other language by initially translating it into Bangla, and then re-translating it into various languages such as English, French, Hindi, and more.
- Inadequate resources and a need for well-documented data are significantly evident for most languages, especially dialectal ones. This dataset can lead to a bigger dataset for both languages. Moreover, this dataset can be advantageous to researchers in the computational linguistic segment.

- ChatgaiyyaAlap can be beneficial for researchers who need to create a sound-based dataset or text-to-speech dialect conversion system for the Chittagonian dialect or standard Bangla language for further research work in this area. Additionally, this dataset can be used for developing a dialect-specific intelligent system.
- ChatgaiyyaAlap can be an effective assistant while studying corpus-based dialectometry and the significant variations of the Chittagonian dialect and standard Bangla.
- While creating this dataset contributors ensured that there was no noise like additional spaces, numbers, emojis, and punctuations in the dataset.

2. Background

Different dialects of the standard Bangla language are spoken in various parts of Bangladesh [1]. The use of a person's dialect is heavily influenced by their socioeconomic status, culture, and the place where they were born and raised. Different Bengali dialects may differ in terms of word choices and sentence structures. Various dialects of the Bangla language are spoken in Bangladesh including Sylheti, Dhakaiya, Rajshahi, Chittagonian, and many others. People in the Sylhet division speak the Sylheti dialect, while those in the Dhaka division speak the Dhakaiya dialect and those in the Chittagong division speak the Chittagonian dialect. Compared to other dialects, the Chittagonian dialect is more difficult for people from different parts of Bangladesh to understand due to its exceptional vocabulary and complex sentence structure. Many words from the Chittagonian dialect are hardly ever found in standard Bangla or any other dialects of Bangla. Table 1 shows the difference between similar meaning words of Chittagonian dialect and standard Bangla.

Table 1
Representation of word differences of standard Bangla and Chittagonian dialect.

Standard Bangla	Chittagonian Dialect
করছি	গইজ্জি
টমেটো	খাট্টাবাইয়্যুন
মুরগি	কুরো
দেওয়ার	দবিরে
কখনো	হনঅত্তে

This diversity in vocabulary makes the Chittagonian dialect complicated and limits the ability to conduct any research work using this dialect. As a result, instead of having large native speakers, research work in the Chittagonian dialect is limited. The primary objective of developing this dataset is to identify the difference between standard Bangla and the Chittagonian dialect. Moreover, we intend to give the Chittangongian dialect a well-documented graphical representation via this dataset. This dataset can be beneficial for researchers who can use this dataset for further studies.

3. Data Description

Language translation is not a recent phenomenon and a handful of research in different languages have already been conducted. Though studies to convert dialects are limited to various languages, dialects of the standard Bangla language are not an exception. In order to convert the Chittagonian dialect into standard Bangla and enhance the proficiency of language models, we developed the dataset "ChatgaiyyaAlap" [4].

The dataset contains two Comma Separated Values (.CSV) files. The initial file contains data categorized into two columns, the first column holds standard Bangla sentences and the second column consists of its corresponding Chittagonian translation. Another file of this dataset, which is our state-of-the-art dictionary file, includes word mappings between standard Bangla and the Chittagonian dialect.

For collecting data in Chittagonian dialect diverse sources like YouTube and Facebook posts, comments, videos, short films, and dramas were explored. Upon completing the data collection and preprocessing, the dataset as well as the dictionary was rigorously evaluated by five professional human evaluators. The five evaluators were male and all were university graduates. The evaluators were originally from Chittagong district in Bangladesh and were living in Dhaka during the experiment. The evaluators were native speakers of the Chittagonian dialect and also had expertise in the standard Bangla language and dialect. The assistance of the evaluators was crucial for finding errors and potential biases. They checked for the accuracy of the dataset along with the quality of translation, which enhanced the data quality and the usefulness of the ChatgaiyyaAlap dataset.

Data collection for the Chittagonian dialect had some significant challenges, while the limitation of resources was the main drawback. To minimize this issue, sentences of standard Bangla from multiple social media sites were gathered. This approach helped to increase the volume of the dataset, though extensive data cleaning and preprocessing were required to erase biases like: spelling mistakes, punctuation, emojis, and numbers.

Like any other language, multiple synonyms for a standard Bangla and Chittagonian dialectal word exist, though we prioritized using more appropriate words in our dataset rather than using alternative synonyms for the same phrase to avoid any misunderstandings [3]. This approach reduced the ambiguity of the dataset and enhanced the translation process. For further enhancement, we have maintained a dictionary file that assisted us in selecting the most suitable Chittagonian word for a standard Bangla word. Consequently, the dataset consists of two files one is Chittagong and Bangla sentences and the other one is a dictionary file.

Fig 1 represents the file structure of the dataset.

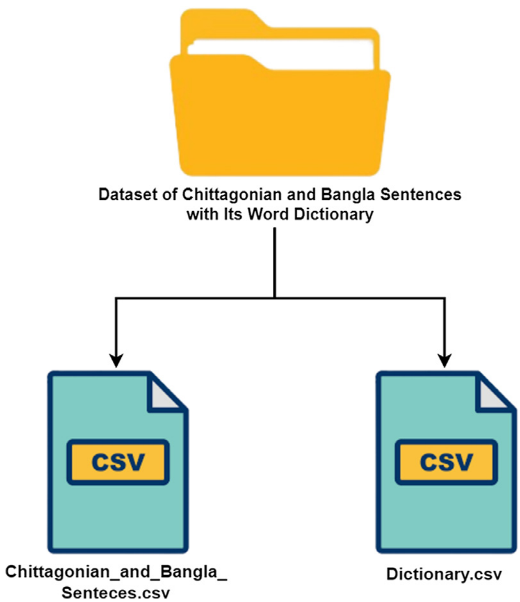

Fig. 1. File structure of the dataset.

4. Experimental Design, Materials and Methods

4.1. Data collection

Predominantly, a wide range of data is gathered from people's posts and comments on social media in text, and video format. There are a few dramas available on different social media platforms where the dialogues are in Chittagonian dialect.

Our first approach was to collect a range of sentences in the Chittagonian dialect and to achieve that, we collected sentences from comments and posts on YouTube and Facebook. After preprocessing, five professional translators helped to convert Chittagonian dialectical sentences into standard Bangla.

In this process, we utilized a dictionary to translate Chittagonian sentences into Standard Bangla. Initially, we added the distinct words to the dictionary, ensuring their spelling was correct. If a word or its synonym was found in the dictionary, we substituted the newly added words with their counterparts from our dictionary. This approach was adopted as we translated the Chittagonian sentences into Standard Bangla. Sentences in Chittagonian dialect and standard Bangla both are assembled into one text file and both types are divided by the tab character in a similar row. However, this process was not rapid for data collection due to the lack of data in the Chittagonian dialect.

To speed up the data collection process and collect more data, we opted for manually collecting data from different dramas and videos available on social media where the dialogues were in the Chittagonian dialect. Converting the video data into text data for the Chittagonian dialect was a delicate matter to tackle. However, it amplified the volume of data and made the data collection smoother. The data from the videos were then converted into standard Bangla with the help of professional translators using the dictionary and stored in the existing text file. But still, the data collection process was steady and to accelerate the process we collected standard Bangla data from Facebook and YouTube comments, posts and after preprocessing it with the help of translators, we converted the data into Chittagonian dialect from standard Bangla.

With all this collection process a volume of 4012 unmatched (unique) sentences were collected during the time of data collection. To ensure the authenticity of the data strict guidelines were followed during the data collection and analysis process. To validate the data, the file was checked and reviewed by multiple native speakers individually and via group discussion of the speakers. They checked the dataset and made modifications if needed, which made the dataset more reliable. Fig. 2 shows the data collection and cleaning process.

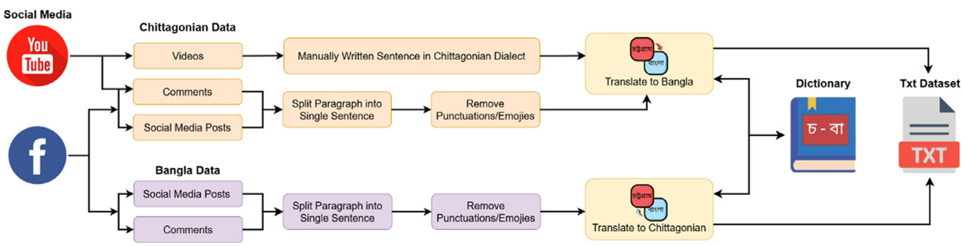

Fig. 2. Dataset preparation methodology.

4.2. Pre-processing

Initially, the data was a collection of several paragraphs but afterward, the paragraphs were segmented by writing a single sentence in a row and the file was in text file format (.txt). After completing our data collection, the entire dataset was converted into .CSV format. To prevent any anomaly the overall dataset was checked manually. Removing noise, for instance numbers, irrelevant blank spaces, emojis, hashtags and typos from the dataset is essential in data preprocessing for developing a clean dataset [2]. So that researchers who are willing to work with the Bangla language or Chittagonian dialect can use this dataset and implement modern artificially intelligent algorithms and encounter any difficulties. To convert any language properly sufficient preprocessed data is a must.

Table 2 showcases the data pre-processing part, while we picked some data to illustrate how data was pre-processed in an elaborate manner. We gathered data from various sources in order to translate data from Chittagonian to standard Bangla. However, we occasionally discovered paragraphs while gathering data. For instance, we found a YouTube comment in Chittagonian dialect: "অ্যাঁই ঘরত আছি হাইতে ফাররি। কন্নিতু মানুষ হানা ন পায় 😔". Primarily, we divided this paragraph into two single sentences and eliminated any unwanted punctuations, emojis, numbers, and spaces from the data. After the paragraph was divided into two single sentences, "অ্যাঁই ঘরত আছি হাইতে ফাররি" and "কন্নিতু মানুষ হানাা ন পায়" was the targeted data for translation. Finally, we converted the data into its standard Bangla format, which is "আমি ঘরে আছি খেতে পারছি "and "কন্নিতু মানুষ খাবার পায় না" with the help of our state-of-the-art dictionary. Identical situations were evident while collecting standard Bangla sentences and to address the issue similar methods were followed.

For the better understanding of readers we have included the comparable English translation of data in Table.2 which does not belong to the ChatgaiyyaAlap dataset.

To translate the data we sought assistance from the experts and with their guidance we were able to develop a state-of-the-art dictionary with time. In the dictionary, we strictly maintained single-word meanings for both standard Bangla and Chittagonian dialect. Any words found unknown were immediately added into the dictionary while collecting data and the effective participation of the professionals facilitated us to find the most appropriate word for translation.

Limitations

Establishing a text dataset containing dialectal language accompanies endless challenges and limitations and this dataset is no different from that. This dataset represents the variation in word selection and sentence structure between standard Bangla and Chittagonian dialect. Due to the limited availability of resources for the Chittagonian dialect, the amount of collected sentences was also limited.

Due to sentence limitations in the dataset, the variety of unique words is also restricted. Only more relevant and widely used words for both Chittagonian dialect and standard Bangla language are used in this dataset, without any synonyms, which leads to a decrease in vocabulary diversity.

Challenges

Collecting data from various resources and cleaning the data was an immense dispute. Data collection is another area where we encountered difficulties because a significant amount of data was assembled from YouTube and Facebook posts and comments. The majority of the posts and comments contain typos, extra spaces, emojis, and numbers that are not relevant to our dataset. Eliminating emojis, numbers, punctuations, additional spaces, and spelling mistakes was a complex issue. The data preprocessing part became more complex due to the wrong structure of Chittagonian sentences and incomplete sentences.

Additionally, the dataset demands manual preprocessing which makes the process labor-intensive and time-consuming. The prevention of using synonymous words and avoiding repetitive sentences are also being checked throughout the time of creating the dataset.

Table 2
Representation of data pre-processing.

Chittagonian to standard Bangla				Standard Bangla to Chittagonian			
Raw Data (Chittagonian Dialect)	Processed data	Translated data	Equivalent English translation	Equivalent English translation	Raw data (standard Bangla)	Processed data	Translated data
অ্যাঁই ঘরত আছি হাইতে ফারির। কিন্তু মানুষ হানা ন পায়😔	অ্যাঁই ঘরত আছি হাইতে ফারির কিন্তু মানুষ হানা ন পায়	আমি ঘরে আছি খেতে পারছি কিন্তু মানুষ খাবার পায় না	I am at home, can eat But people don't get food	Where should I go during the rainy season? How will I survive when there is a storm like this? 😭	বর্ষা কালে কোথায় যাবো? এমন করে ঝড় হলে কেমন করে থাকবো 😭	বর্ষা কালে কোথায় যাবো এমন করে ঝড় হলে কেমন করে থাকবো	বর্ষা কাইল্যা হডে যাইয়ুম এন গরি ঝড় অইলে ক্যান গরি থাইক্কুম
অ্যাঁর তোঁয়ারে বেশি গম লার😉	অ্যাঁর তোঁয়ারে বেশি গম লার	আমার তোমাকে বেশি ভাল লাগে	I like you more	You don't get hanged for stealing bananas!!!	কলা চুরির জন্য ও ফাসি হয় না!!!	কলা চুরির জন্য ও ফাসি হয় না	কলো চুরির লাই ও ফাসি ন অয়
বেজ্ঞুনে অ্যাঁর সামনে তুন চলি যাইবেন	বেজ্ঞুনে অ্যাঁর সামনে তুন চলি যাইবেন	সবাই আমার সামনে থেকে চলে যাবেন	Everyone will leave in front of me	When you see a poor person, you spit😤	গরিব দেখলে থুতু ফেলে😤	গরিব দেখলে থুতু ফেলে	গরিব চ্যাঁইলে ছেপ ফেলায়

Ethics Statement

To build this dataset no human or animal experimentation was performed. The data was collected from different social media platforms and the user's identity is completely anonymized.

CRediT Author Statement

Sinthia Chowdhury: Conceptualization, Data Collection, Data Curation, Validation, Writing – original draft, Resources. **Deawan Rakin Ahamed Remal:** Data Curation, Methodology, Data Collection, Resources, Formal Analysis, Writing- review & editing. **Syed Tangim Pasha:** Investigation, Writing- review & editing. **Ashraful Islam:** Writing- review, Supervision, Project Administration. **Sheak Rashed Haider Noori:** Supervision

Data availability

ChatgaiyyaAlap: A Dataset for Conversion from Chittagonian Dialect to Standard Bangla (Original data) (Mendeley Data).

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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